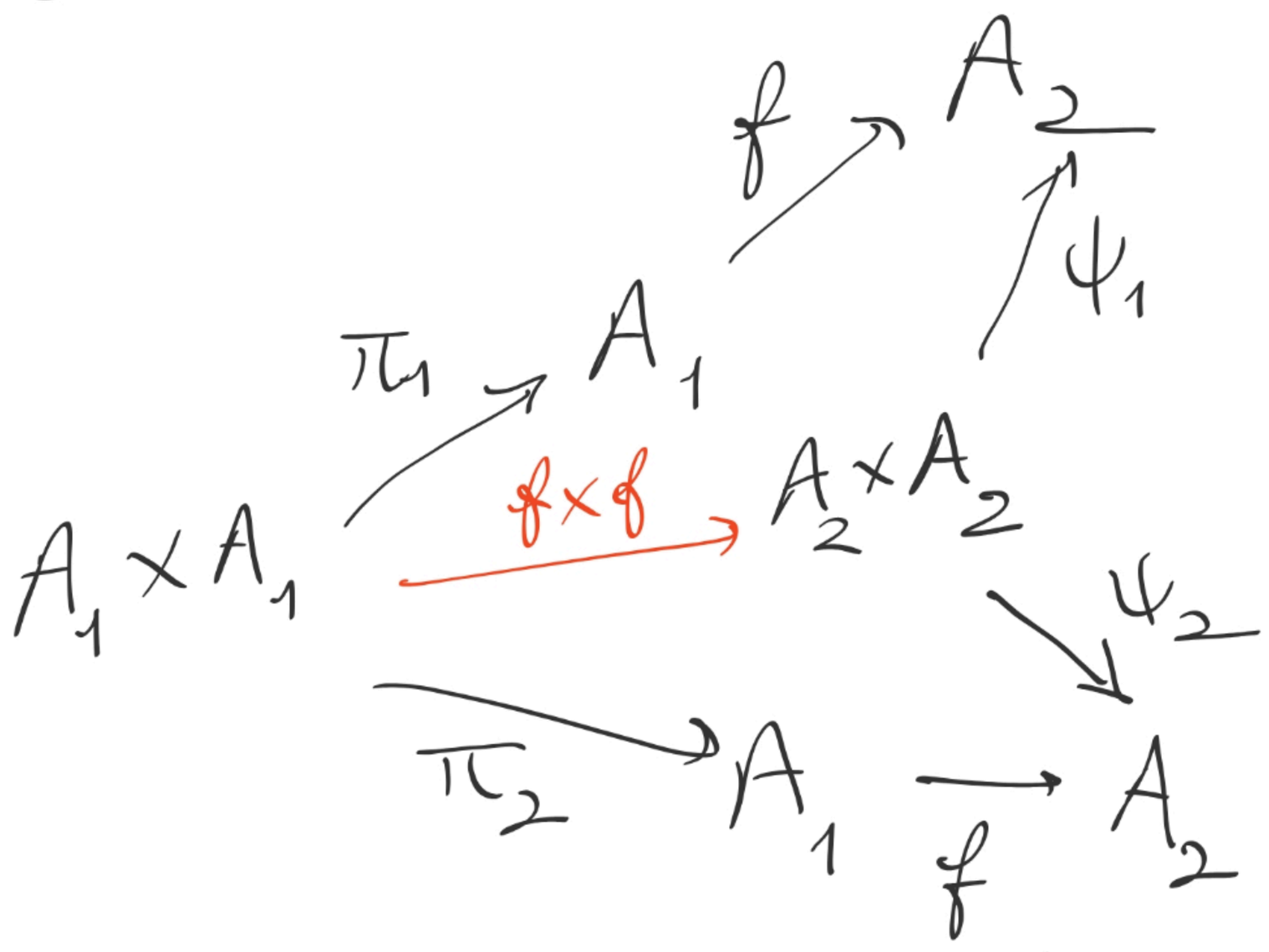
Algebra Monsoon 2026 Exercise : the product of two mappings

Let $f : A_1 \longrightarrow A_2$ be a mapping. Consider the products $(A_1 \times A_1, \pi_1, \pi_2)$ and $(A_2 \times A_2, \psi_1, \psi_2)$.

Consider the mappings (in black)



By the universal property of the product $A_2 \times A_2$, there is a unique mapping $f \times f : A_1 \times A_1 \longrightarrow A_2 \times A_2$ such that $f \circ \pi_1 = \psi_1 \circ (f \times f)$

and $f \circ \pi_2 = \psi_2 \circ (f \times f)$.

More generally, if

$$f : A \longrightarrow B$$

and $g : C \longrightarrow D$

are mappings, then there is a unique mapping

$$f \times g : A \times C \longrightarrow B \times D$$

making the following diagrams commute:

